

Titanic Case study

```
In [ ]: 1 # replace 'male' with 1 and 'female' with 0
2 1. using userdefined function + apply
3 2. using lambda + apply
4 3. using map + dictionary
5 4. using map + lambda
6 5. using replace
7
```

```
In [19]: 1 import pandas as pd
2
3 df=pd.read_csv(r"C:\Users\shubh\Downloads\Data\Data\Titanic.csv")
4 df.head()
```

```
Out[19]:
```

	Unnamed: 0	PassengerId	Pclass	Name	Gender	Age	SibSp	Parch	Ticket	Fare	C
0	0	892	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	
1	1	893	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000	
2	2	894	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	
3	3	895	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	
4	4	896	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	

```
In [5]: 1 df["Gender"].unique()
```

```
Out[5]: array(['male', 'female'], dtype=object)
```

1. using userdefined function + apply

```
In [13]: 1 def userdef(n):
2         if n=="male":
3             return 1
4         else:
5             return 0
```

```
In [21]: 1 df["Gender1"]=df["Gender"].apply(userdef)
          2 df.head()
```

```
Out[21]:
```

	Unnamed: 0	PassengerId	Pclass	Name	Gender	Age	SibSp	Parch	Ticket	Fare	C
0	0	892	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	
1	1	893	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000	
2	2	894	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	
3	3	895	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	
4	4	896	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	

2.using lambda + apply

```
In [24]: 1 df["Gender2"]=df["Gender"].apply(lambda x: 1 if x=="male" else 0)
          2 df.head()
```

```
Out[24]:
```

	Unnamed: 0	PassengerId	Pclass	Name	Gender	Age	SibSp	Parch	Ticket	Fare	C
0	0	892	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	
1	1	893	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000	
2	2	894	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	
3	3	895	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	
4	4	896	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	

3. using map + dictionary

```
In [27]: 1 df["Gender3"]=df["Gender"].map({"male":1,
2                                     "female":0})
3 df.head()
```

```
Out[27]:
```

	Unnamed: 0	PassengerId	Pclass	Name	Gender	Age	SibSp	Parch	Ticket	Fare	C
0	0	892	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	
1	1	893	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000	
2	2	894	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	
3	3	895	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	
4	4	896	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	

4. using map + lambda

```
In [30]: 1 df["Gender4"]=df["Gender"].map(lambda x:1 if x=="male" else 0)
2 df.head()
```

```
Out[30]:
```

	Unnamed: 0	PassengerId	Pclass	Name	Gender	Age	SibSp	Parch	Ticket	Fare	C
0	0	892	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	
1	1	893	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000	
2	2	894	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	
3	3	895	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	
4	4	896	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	

5. using replace

```
In [33]: 1 df["Gender5"]=df["Gender"].replace(['male', 'female'],[1,0])
          2 df.head()
```

```
Out[33]:
```

	Unnamed: 0	PassengerId	Pclass	Name	Gender	Age	SibSp	Parch	Ticket	Fare	C
0	0	892	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	
1	1	893	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000	
2	2	894	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	
3	3	895	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	
4	4	896	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	



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In [ ]:
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In [ ]:
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In [ ]:
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